Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 06-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Placebo-controlled Clinical Study to Evaluate Mavacamten in Adults With Symptomatic Non-obstructive Hypertrophic Cardiomyopathy **Short Title/Acronym:** ODYSSEY-HCM **Protocol Number:** CV027-031 / 2021-005329-26 **NCT Number:** NCT05582395 **Posting ID:** 117788253390471304 **Sponsor Name / Company:** Bristol Myers Squibb (MyoKardia, Inc.) **Investigational Product:** Mavacamten (BMS-986427) **Trade Name:** Camzyos **Phase of Development:** Phase 3 **Trial Status:** Completed **Medical Monitor:** GSM-CT Representative (Bristol Myers Squibb)

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and efficacy of mavacamten compared with placebo in participants with symptomatic non-obstructive hypertrophic cardiomyopathy (nHCM). **Secondary Objectives:** To evaluate the change from baseline in ventilatory efficiency (VE/VCO2) slope, New York Heart Association (NYHA) functional class, and N-terminal pro B-type natriuretic peptide (NT-proBNP).

2.2 Background & Rationale Mavacamten is a first-in-class cardiac myosin inhibitor approved for obstructive hypertrophic cardiomyopathy (oHCM). However, its effects in non-obstructive HCM (nHCM) remain uncertain. The ODYSSEY-HCM trial was conducted to test the hypothesis of whether a cardiac myosin inhibitor would improve measures of feel, functional capacity, and symptoms for patients with nHCM, addressing a disease area with a significant unmet need for targeted therapeutic options.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Participant, Investigator) **Randomization:** Randomized (1:1 ratio), Parallel Assignment

3.2 Planned Enrollment & Duration Target N: 580 adult patients **Number of Sites:** Approximately 180 sites across 21 countries worldwide (Americas, Europe, Middle East, and Asia) **Estimated Study Start:** 2022 **Study Completion:** Week 48 for primary endpoint evaluation (Completed)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age > 18 years. 2. Diagnosis of HCM consistent with ACCF/AHA/ESC guidelines: unexplained left-ventricular hypertrophy with non-dilated ventricular chambers and maximal left ventricular (LV) wall thickness ≥ 15 mm (or ≥ 13 mm with positive family history of HCM). 3. Peak left ventricular outflow tract (LVOT) pressure gradient < 30 mm Hg at rest and < 50 mm Hg with provocation. 4. New York Heart Association (NYHA) Class II or III.

4.2 Exclusion Criteria 1. Known infiltrative or storage disorder causing cardiac hypertrophy that mimics nHCM (e.g., Fabry disease, amyloidosis, or Noonan syndrome). 2. History of unexplained syncope within 6 months prior to screening. 3. History of sustained ventricular tachyarrhythmia (> 30 seconds) within 6 months prior to screening.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Mavacamten (starting at 5 mg per day and adjusted up to a maximum of 15 mg per day based on left ventricular ejection fraction) or matching placebo with sham dose adjustment. **ATC Code:** C01EB24 **EMA URL:** https://www.ema.europa.eu/en/documents/product-information/camzyos-epar-product-information_en.pdf **Route of Administration:** Oral (Capsule) **Duration of Treatment:** 48 weeks for the primary outcome measure, with total participation lasting between 52 and 134 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Standard background medical therapies for heart failure and HCM co-morbidities as prescribed by the treating physician. **Prohibited:** Concurrent medications that strongly induce or inhibit CYP450 enzymes that heavily alter mavacamten plasma concentrations, or therapies carrying a severe risk of exacerbating systolic dysfunction.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Echocardiogram, CPET
Baseline	Day 0	Randomization, First Dose of Mavacamten/Placebo
Interim	Weeks 4 to 36	Safety Labs, Echocardiogram for LVEF monitoring and dose adjustment
End of Study	Week 48	Final CPET (pVO2 assessment), KCCQ-23 Questionnaire, Safety Labs

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Dual-primary endpoints examining changes from baseline to Week 48 in the Kansas City Cardiomyopathy Questionnaire Clinical Summary Score (KCCQ-23 CSS) and peak oxygen consumption (pVO2). **Measurement Method:** Patient-reported outcome questionnaire (KCCQ-23) and Cardiopulmonary Exercise Testing (CPET).

7.2 Secondary & Exploratory Endpoints 1. Change from baseline in ventilatory efficiency (VE/VCO2) slope to Week 48. 2. Proportion of participants with at least 1 class of New York Heart Association (NYHA) improvement from baseline to Week 48. 3. Change from baseline in N-terminal pro B-type natriuretic peptide (NT-proBNP) to Week 48.

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine patient assessments and echocardiography monitoring, focusing closely on reductions in left ventricular ejection fraction (LVEF) and potential heart failure due to systolic dysfunction. **Serious Adverse Events (SAEs):** Standard reporting timelines (within 24 hours of site awareness). **Data Monitoring Committee (DMC):** External independent DMC to assess unblinded safety data and LVEF metrics during the trial.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation. Safety Set for safety metrics. **Sample Size Justification:** Total $N=580$ uniformly randomized (1:1) to adequately power dual primary endpoints. **Handling of Missing Data:** Multiple Imputation (MI) and standard robust generalized linear modeling for continuous endpoints like pVO2 and KCCQ-CSS scores.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of trial sites to ensure protocol adherence, especially regarding echocardiogram-guided dose adjustments. **Audit Trail:** eCRF locking, centralized reading for core laboratory interpretations (e.g., CPET, Echo), and prompt data clarification.

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11. Study Identifiers & Governance

Primary ID: CV027-031 **Registry Number:** NCT05582395 **Posting ID:** 117788253390471304 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of Mavacamten in Non-Obstructive Hypertrophic Cardiomyopathy **Official Regulatory Title:** A Randomized, Double-blind, Placebo-controlled Clinical Study to Evaluate Mavacamten in Adults With Symptomatic Non-obstructive Hypertrophic Cardiomyopathy

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12. Clinical Framework (Cardiomyopathy Specific)

Primary Condition / Preferred UMLS Name: Non-Obstructive Hypertrophic Cardiomyopathy (nHCM) **Disease Areas:** Cardiomyopathy, Hypertrophic **SNOMED ID:** 56246009 **Diagnosis Threshold:** LV wall thickness ≥ 15 mm and Peak LVOT pressure gradient < 30 mm Hg at rest **Medical Methodology:** Cardiac Myosin Inhibition **Optimization Target:** Improvement of functional capacity (pVO2) and symptomatic burden (KCCQ-23)

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13. Study Procedures & Methodology

Management Pathway: Echocardiogram-guided dose adjustment of mavacamten **Education Protocol (HCP):** Site investigator training on protocol-specific LVEF criteria and dose-titration boundaries. **Education Protocol (Patient):** Symptom monitoring, CPET preparation, and reporting of shortness of breath/fatigue. **Quality Audit Mechanism:** Core laboratory independent reviews for echocardiograms and exercise testing.

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14. Observation & Follow-Up Schedule

Total Duration: 48 weeks for the primary endpoint; overall participation up to 134 weeks **Onsite Visit Cadence:** Frequent early visits for dose titration, followed by standard interval visits (e.g., Weeks 12, 24, 36, 48) **Remote Monitoring Frequency:** Routine data monitoring/electronic patient-reported outcomes **Checklist Review Interval:** Correlated to onsite echocardiography assessments for dose scaling.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]